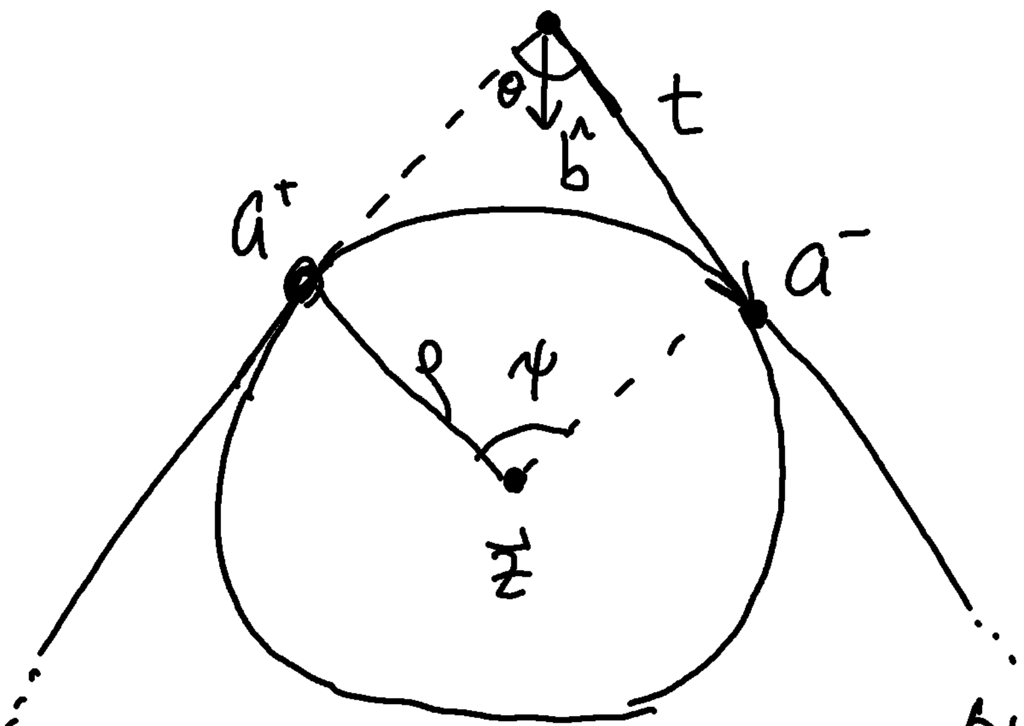
Rounded Definitions



$$\boxed{\begin{aligned}\vec{a}^- &= \vec{V}_i - t\,\hat{e}_{z'(i)} \\ \vec{a}^+ &= \vec{V}_i + t\,\hat{e}_i\end{aligned}}$$

$$\tan \theta/2 = \frac{\rho}{t}$$

$$\boxed{t = \rho / \tan \theta/2}$$

$$\sin \theta/2 = \frac{\rho}{|\vec{V}_i - \vec{z}|}$$

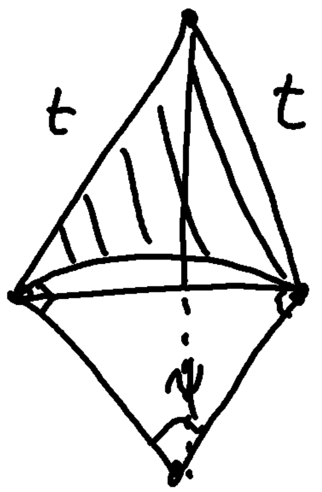
so

$$\boxed{\vec{z} = \vec{V}_i + \left(\frac{\rho}{\sin \theta/2}\right)\hat{b}}.$$

$$\boxed{\theta = \arccos\left(-\hat{e}_{z'(i)} \cdot \hat{e}_i\right)}$$

$$\boxed{\hat{b} = \frac{\hat{e}_i - \hat{e}_{z'(i)}}{|\hat{e}_i - \hat{e}_{z'(i)}|}}$$

$$\boxed{\psi = \pi - \theta}.$$



$$\boxed{\Delta\rho = -2t + \rho\psi.}$$

Find the area of the backbone now with a line from a to a' and add only the circular segment.

$$\Delta A = \frac{1}{2}\psi\rho^2 - \frac{1}{2}\rho^2 \sin\psi$$

$$\boxed{\Delta A = \frac{1}{2}\rho^2(\psi - \sin\psi).}$$

Now the gradient

$$\frac{\partial \theta_k}{\partial V_j^\alpha} = \frac{\partial}{\partial V_j^\alpha} \arccos\left[-\hat{e}_{z'(k)} \cdot \hat{e}_k \right]$$

$$= \frac{(\delta_{kj} - \delta_{z'(k)j})\left(\frac{\hat{e}_k^\alpha - (\hat{e}_k \cdot \hat{e}_{z'(k)}) e_{z'(k)}^\alpha}{|\vec{e}_{z'(k)}|} \right)}{\sin\theta_k}$$

$$+ \frac{(\delta_{z'(k)j} - \delta_{kj})\left(\frac{\hat{e}_{z'(k)}^\alpha - (\hat{e}_k \cdot \hat{e}_{z'(k)}) \hat{e}_k^\alpha}{|\vec{e}_k|} \right)}{\sin\theta_k}$$

$$\boxed{\frac{\partial \theta_k}{\partial V_j^\alpha} = \frac{(\delta_{kj} - \delta_{z'(k)j})}{\sin\theta_k}\left(\frac{\hat{e}_k^\alpha - \cos\theta_k \, \hat{e}_{z'(k)}^\alpha}{|\vec{e}_{z'(k)}|} \right) + \frac{(\delta_{z(k)} - \delta_{kj})}{\sin\theta_k}\left(\frac{\hat{e}_{z'(k)}^\alpha - \cos\theta_k \, \hat{e}_k^\alpha}{|\vec{e}_k|} \right)}$$

$$P = P_{backbone} - \sum_{j=1}^{n}\left(2t_j - \rho\psi_j\right)$$

$$\frac{\partial P}{\partial V_k^\alpha} = \sum_{j=1}^{n} \frac{\partial}{\partial V_k^\alpha}\left[\left|\vec{V}_{z(j)} - \vec{V}_j\right| - 2t_j + \rho\psi_j\right]$$

$$= \sum_{j=1}^{n}\left[\hat{e}_j^\alpha\left(\delta_{z(j)k} - \delta_{jk}\right) - 2\frac{\partial}{\partial V_k^\alpha}\frac{\rho}{\tan\theta_j/2} - \rho\frac{\partial\theta_j}{\partial V_k^\alpha}\right]$$

$$\frac{\partial P}{\partial V_k^\alpha} = \hat{e}_{z'(k)}^\alpha - \hat{e}_k^\alpha + \rho\sum_{j=1}^{n}\frac{\partial\theta_j}{\partial V_k^\alpha}\left(-1 + \csc^2\theta_j/2\right)$$

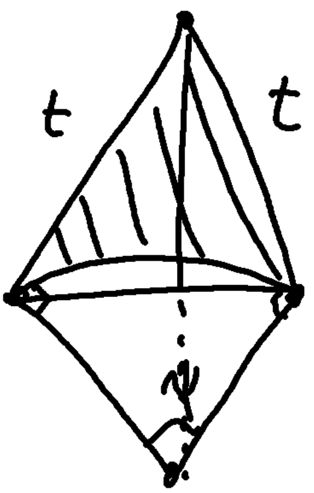
$$\boxed{\frac{\partial P}{\partial V_k^\alpha} = \hat{e}_{z'(k)}^\alpha - \hat{e}_k^\alpha + \rho\sum_{i=1}^{n}\frac{\partial\theta_i}{\partial V_k^\alpha}\cot^2\theta_i/2 .}$$

Next we separate this into a kite, wedge, and polygon

convexity

$$A = A_{backbone} - \sum_{j=1}^{n} \left(A_{kite,j} - A_{wedge}\right) S_j$$

$$= \sum_{j=1}^{n} \left[\frac{1}{2} \epsilon^{\alpha}{}_{\beta}\, e_j^{\beta}\, e_{z(j)}^{\alpha} - A_{kite,j} + \frac{\psi_j}{2} \rho^2 \right]$$



$$A_{kite} = \frac{1}{2}\left(2t \sin \theta/2\right)\left(\rho \cos \psi/2 + t \cos \theta/2\right)$$

$$= t \cancel{\sin} \theta/2 \left(\rho \frac{\sin^2 \theta/2 + \cos^2 \theta/2}{\cancel{\sin \theta/2}}\right)$$

$$A_{kite} = t\rho$$

$$\boxed{A_{kite} = \frac{\rho^2}{\tan \theta/2}}$$

$$\boxed{A_{kite} - A_{wedge} = \rho^2 \left(\cot \theta/2 - \psi/2\right)}$$

$$\boxed{\frac{\partial A}{\partial V_k{}^{\alpha}} = \frac{1}{2} \epsilon^{\alpha}{}_{\beta} \left(\vec{V}_{z(k)} - \vec{V}_{z'(k)}\right)^{\beta} + \frac{1}{2} \rho^2 \sum_{i=1}^{n} S_i \cot^2\!\left(\frac{\theta_i}{2}\right) \frac{\partial \theta_i}{\partial V_k{}^{\alpha}}.}$$

from $1 - \csc^2$

Intersections

Let $\boxed{\vec{P}_k(s) = \vec{a}_k^{\,+} + s\left(\vec{a}_{z(k)}^{\,-} - \vec{a}_k^{\,+}\right)}$

be a straight segment where $s \in [0,1]$.

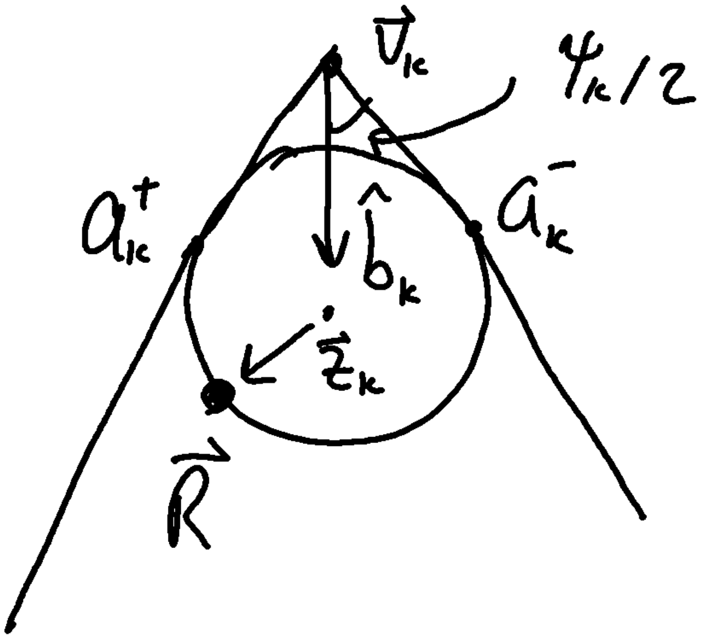
Points on the arc have the form

$$\boxed{\vec{Q}_k(\varphi) = \vec{Z}_k + \rho_k \begin{pmatrix} \cos\varphi \\ \sin\varphi \end{pmatrix}}$$

and the arc is defined for

$$\boxed{\varphi_k^{\pm} = \arctan\left(\vec{a}_k^{\,\pm} - \vec{Z}_k\right).}$$

and spans $\psi_k = \varphi_k^+ - \varphi_k^-$.



For a point $\vec{R}$ on the circle,

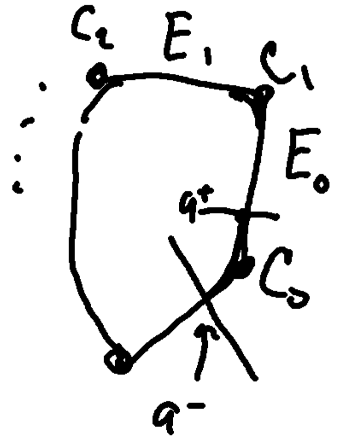
$\vec{R}$ lies on the arc iff

$$\boxed{\frac{\vec{R}-\vec{z}_k}{\rho_k}\cdot\hat{b}_k \leq -\cos\psi_k/2} \quad (AVT)$$

Coin this AVT for arc-validity test.

Ordering features

We order the features of a polygon like



So there are $2n$ features per polygon

The crossing value is (where A and B are the intersecting pair)

$$\boxed{O_A = f + \begin{cases}\lambda & \text{if } f \text{ even}\\ s & \text{if } f \text{ odd.}\end{cases}}$$

Where λ is the fractional part $\lambda \in [0,1)$ of the arc. For an arc $k = f/2$, we have

$$\boxed{\lambda \equiv \frac{a - a_k^-}{\psi_k}.}$$

Since $a_k^+ - a_k^- = \psi_k$

We need to find a (where the intersection occurs) and S.

<u>finding S</u>

S depends on the sort of intersection you have.

If you have two edges, then

$$\boxed{\begin{aligned} &\vec{P}_k(s) = \vec{a}_k^{\,+} + s\vec{u}_k \,, \quad \vec{u}_k = \vec{a}_{z(k)} - \vec{a}_k^{\,+} \\ &\vec{P}_\ell(t) = \vec{a}_\ell^{\,+} + t\,\vec{u}_\ell \end{aligned}}$$

if we let $\vec{w}_{k\ell} = \vec{a}_\ell^{\,+} - \vec{a}_k^{\,+}$ then

$\vec{P}_\ell(t) - \vec{P}_k(s) = \vec{w}_{k\ell} - s\vec{u}_k + t\vec{u}_\ell = \vec{0}$ for an intersection. Therefore

$$-s u_k^x + t u_\ell^x + w_{k\ell}^x = 0$$

$$t = \frac{s u_k^x - w_{k\ell}^x}{u_\ell^x}$$

$$-s\,u_k^{y} + \left(\frac{s\,u_k^{x} - w_{kl}^{x}}{u_l^{x}}\right) u_l^{y} + w_{kl}^{y} = 0$$

$$s\left(-u_k^{y} + \frac{u_k^{x}}{u_l^{x}}\,u_l^{y}\right) = \frac{w_{kl}^{x}}{u_l^{x}}\,u_l^{y} - w_{kl}^{y}$$

$$s = \frac{w_{kl}^{x}\,u_l^{y} - w_{kl}^{y}\,u_l^{x}}{-u_k^{y}\,u_l^{x} + u_k^{x}\,u_l^{y}}$$

$$\boxed{s = \frac{\varepsilon_{\alpha\beta}\,w_{kl}^{\alpha}\,u_l^{\beta}}{\varepsilon_{\alpha\beta}\,u_k^{\alpha}\,u_l^{\beta}}.}$$

If the edge intersects an arch then $\vec{P}_k(s)$ intersects a point $\vec{R}_l$ on the full circle associated with arc l. Then $|\vec{R}_l - \vec{z}_l| = \rho$

If we then let $\vec{w}_{kl} = \vec{a}_k^{\,+} - \vec{z}_l$ where the segment starts at point $\vec{a}_k^{\,+}$, then we can proceed.

this gives $\vec{R}_\ell = \vec{P}_k(s)$ so that

$$|\vec{R}_\ell - \vec{z}_\ell|^2 = |\vec{P}_k(s) - \vec{z}_\ell|^2$$

$$\rho^2 = |\vec{w}_{k\ell} + s\vec{u}_k|^2$$

$$= |\vec{w}_{k\ell}|^2 + 2s(\vec{w}_{k\ell}\cdot\vec{u}_k) + s^2|\vec{u}_k|^2$$

Then

$$0 = \left(\frac{|\vec{w}_{k\ell}|}{|\vec{u}_k|}\right)^2 + 2s\,\frac{\vec{w}_{k\ell}\cdot\vec{u}_k}{|\vec{u}_k|^2} + s^2 - \rho^2$$

$$s_\pm = -\frac{\vec{w}_{k\ell}\cdot\vec{u}_k}{|\vec{u}_k|^2} \pm \frac{1}{|\vec{u}_k|^2}\sqrt{(\vec{w}_{k\ell}\cdot\vec{u}_k)^2 + |\vec{u}_k|^2(\rho^2 - |\vec{w}_{k\ell}|^2)}$$

$$\boxed{\text{Let } D_{k\ell} = (\vec{w}_{k\ell}\cdot\vec{u}_k)^2 + |u_k|^2(\rho^2 - |w_{k\ell}|^2)}$$

$$\boxed{s_\pm = \frac{-(\vec{w}_{k\ell}\cdot\vec{u}_k) \pm \sqrt{D_{k\ell}}}{|\vec{u}_k|^2}}$$

)Keep each root if $s_\pm$ is valid (between 0 and 1 and passes AVT).

It could be that an edge intersects an arc twice.

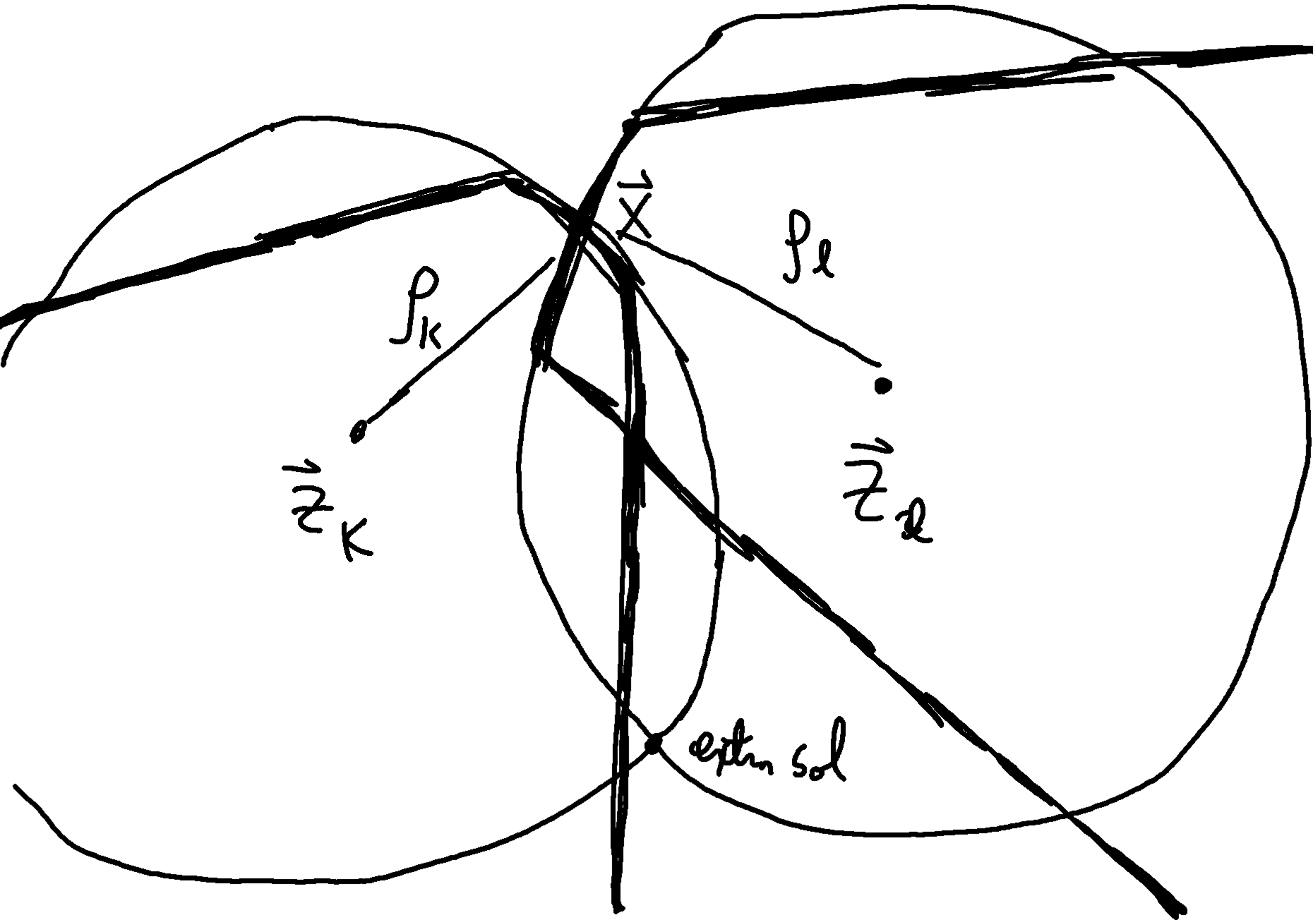
To get the corresponding angle α, we simply find $\vec{P}_k(s)$ and then have

$$\boxed{\alpha = \arccos\left(-\frac{\left(\vec{P}_k(s) - \vec{Z}_k\right)\cdot \hat{b}_k}{\rho_k} \right).}$$

Arc - Arc intersection

We consider two arcs colliding at point $\vec{X}$ so that $|\vec{X} - \vec{Z}_K|^2 = \rho_K^2$ and $|\vec{X} - \vec{Z}_\ell|^2 = \rho_\ell^2$

These solutions should lie on a line



Solving,

$$|\vec{x}|^2 - 2\vec{x}\cdot\vec{z}_k + |\vec{z}_k|^2 = \rho_k^2$$

$$|\vec{x}|^2 - 2\vec{x}\cdot\vec{z}_\ell + |z_\ell|^2 = \rho_\ell^2$$

$$2\vec{x}\cdot(\vec{z}_\ell - \vec{z}_k) + |\vec{z}_k|^2 - |\vec{z}_\ell|^2 = \rho_k^2 - \rho_\ell^2$$

if we let $\vec{n}_{k\ell} = \vec{z}_\ell - \vec{z}_k$ and

$\hat{n}_{k\ell} = \dfrac{\vec{n}_{k\ell}}{|\vec{n}_{k\ell}|}$ then